

Tata iQ Online Retail Analysis

Executive Summary | SQL + Power BI | April 2026

Project Context

Simulation	Tata iQ Data Visualisation — Forage
Stakeholders	CEO and CMO of an Online Retail Business
Dataset	Online Retail — Forage (Dec 2010 – Dec 2011)
Raw Rows	541,909
Clean Rows (SQL)	530,100 Clean Rows (Power BI): 227,703
Tools	PostgreSQL via pgAdmin + Power BI Desktop

Key Numbers at a Glance

£285,446	Netherlands Top Int'l Revenue	£1,509,496	November 2011 Peak Month	£280,206	Customer 14646 Top Customer
34.42%	One-Time Buyers Retention Risk	69.82%	Top 5 Markets Int'l Revenue Share	£3,303,268	Q4 2011 Strongest Quarter

SQL Analysis — PostgreSQL

Data Setup and Cleaning

- All 541,909 rows imported as TEXT columns to prevent type errors during import
- SET datestyle = 'MDY' used to correctly parse MM/DD/YY date format before ALTER TABLE
- Column types converted: Quantity to INTEGER, UnitPrice to NUMERIC(10,2), InvoiceDate to TIMESTAMP
- Clean table created filtering out returns (quantity < 1), pricing errors (unit_price <= 0), and blank descriptions
- Revenue column added as quantity × unit_price — final clean dataset: 530,100 rows

Data Validation Results

- 530,100 rows confirmed in clean table — all invoice_no and description columns fully populated
- 132,220 rows have no Customer ID (25% of data) — customer-level analysis covers 397,880 rows only
- Min quantity = 1, min unit_price = £0.04 — no negatives slipped through cleaning
- Max quantity = 80,995 and max unit_price = £13,541.33 — extreme outliers present, documented as limitations

10 Analysis Queries — Summary of Findings

Q1 — Revenue by Country	Netherlands leads at £285,446 (94 orders). EIRE second at £283,454 (288 orders). Top 5: Netherlands, EIRE, Germany, France, Australia.
Q2 — Monthly Trend 2011	November peaks at £1,509,496. Strong Q4 build-up Sep–Nov. December at £638,793 is incomplete data — not a seasonal decline.
Q3 — Top 10 Customers	Customer 14646 = £280,206 across 73 orders. Customer 16446 = £168,473 in just 2 orders. Heavy revenue concentration risk.
Q4 — Top 10 Products	DOTCOM POSTAGE ranks first at £206,249 — shipping charge, not a product. REGENCY CAKESTAND 3 TIER = £174,485 — top real product.

Q5 — Repeat Customers	Customer 12748 placed 209 orders — most frequent buyer across full 13-month period. High order count ≠ high revenue always.
Q6 — Avg Order Value	Singapore tops AOV at £3,040 per order (7 orders). Netherlands second at £3,037. Australia at £2,430 with 57 orders — growth market.
Q7 — Revenue by Quarter	Q4 2011 = £3,303,268. Consistent quarterly growth: Q1 £1.9M → Q2 £2.1M → Q3 £2.5M → Q4 £3.3M. Clear seasonality.
Q8 — Frequency Segments	45.48% occasional buyers (2–5 orders). 34.42% one-time buyers — retention risk. Only 7.77% loyal buyers (11+ orders).
Q9 — Quantity Demanded	Netherlands tops at 200,361 units (94 orders). EIRE second at 147,173 units. Sweden bulk volume buyer but lower revenue per unit.
Q10 — Revenue Contribution	Netherlands 17.39% + EIRE 17.27% = 34.66% combined. Top 5 markets = 69.82% of international revenue cumulatively.

Power BI Dashboard

Data Cleaning and Model

- InvoiceDate fixed using Change Type with Locale (English UK) — mixed format resolved
- Filtered Quantity >= 1 and UnitPrice > 0 — clean dataset: 227,703 rows
- Revenue column created as Quantity × UnitPrice in Power Query
- Date_Table created using CALENDAR() DAX function, marked as Date Table
- KPI_Measures table created via Enter Data — all DAX measures stored centrally
- One-to-many relationship: Date_Table[Date] → Online Retail Data Set[InvoiceDate]

Dashboard Pages

Page 1 — Monthly Revenue Trend (CEO)	Line chart showing month-by-month revenue for 2011. Peak in November at £1,509,496. Strong Q4 pattern visible. December drop flagged as incomplete data.
Page 2 — Top 10 Countries (CMO)	Clustered bar chart of revenue + quantity by country, excluding UK. EIRE leads at £102K and 54K units. European markets dominate international performance.
Page 3 — Top 10 Customers (CMO)	Clustered bar chart showing top customers by revenue, excluding blank Customer IDs. Customer 14596 at £168K is highest. Top 2 combined exceed £300K.
Page 4 — Global Demand (CEO)	Treemap showing quantity by country excluding UK. Netherlands and EIRE are largest boxes. Gradient color from dark navy (low) to sky blue (high demand).

Data Quality and Limitations

- SQL retained 530,100 rows vs Power BI's 227,703 — PostgreSQL date parsing handled formats Power BI could not. Difference is intentional and documented.
- 132,220 rows (25%) have no Customer ID — all customer findings apply only to identified buyers
- DOTCOM POSTAGE and POSTAGE entries inflate product revenue — must be excluded in product analysis
- PAPER CRAFT LITTLE BIRDIE shows £168,470 from 1 order — likely a bulk entry error, treated as an outlier
- Dataset covers only 13 months — year-on-year comparisons are not possible
- No cost or margin data available — profitability analysis cannot be performed

Recommendations

Prioritise Netherlands and EIRE	Together they contribute 34.66% of international revenue. Assign dedicated account managers and protect these relationships.
Plan for Q4 demand surge	Revenue grows from £1.9M in Q1 to £3.3M in Q4. Inventory and logistics planning must begin in Q2 for peak season readiness.
Address 34.42% one-time buyer rate	Over one third of customers never returned. A post-purchase follow-up or loyalty incentive programme should be tested immediately.
Exclude postage from product reporting	DOTCOM POSTAGE appears as top revenue item. Filtering it out will give accurate product performance visibility.
Explore Australia as a growth market	High AOV (£2,430) with 57 orders — decent frequency and high value. Targeted marketing investment recommended.